DASC 32003: Optimization Methods in Data Science
Course Project

Deadline: April 17

Intermediate Deliverable Deadline: March 13

1 Overview

Students will work in groups of three on this assignment. This assignment is worth 100% of the overall project grade.

2 Purpose

The purpose of this assignment is to solve a realistic problem by applying optimization. In order to be successful on this assignment, each team must identify a problem that can be solved using optimization, formulate an optimization model to solve the problem, gather data to populate the optimization model, and then use optimization solvers in order to perform analysis of the problem.

3 Identify a Problem

Select an optimization problem that will be the focus of your project. The following list contains some ideas that may help in identifying a topic for your project:

A. Choose a problem related to your DASC concentration. For example:

   If you are pursuing the geospatial data analytics, operations analytics, or supply chain analytics concentration, you might seek to locate a collection of facilities (you choose which type) to provide service to geographically dispersed areas and/or population.

   If you are pursuing the business data analytics or accounting analytics concentration, you might seek to optimize the allocation of funds in an investment portfolio.

B. Choose an established problem and/or application in the operations research literature. For recent applications, I suggest taking a look at some of the papers recently published in the *INFORMS Journal on Applied Analytics*. You could also consider looking in the other INFORMS Journals if you'd like.

C. Choose an idea based on your own interests or hobbies. For some ideas, take a look at some recent projects that were completed by students at the Ohio State University (see https://u.osu.edu/davanloo/teaching/). Alternatively, if you are doing analytics for one of the sports teams on campus, consider solving an optimization problem related to sports analytics.

If your idea is based on someone else's work, you must clearly cite it. In addition, your project must go beyond simply reproducing what someone else's work has already done. I have provided the following guidelines to clarify:

- You may (with citation) use someone else's work as the motivation for your project.

- You may use (with citation) an optimization model that is published in the peer-reviewed academic literature; otherwise, you must formulate your own optimization model.

- You may not (under any circumstances) use someone else's implementation of an optimization model. You are required to implement the model yourself.

- You may not choose an optimization model that is covered in a course you are taking or have taken.

To enable the greatest opportunity for meaningful work on the project, please ask yourself the following questions before settling on a project topic:

- Is it possible to solve your problem by simple analysis (e.g., by performing a few calculations in excel) without using optimization? Do you already know what an optimal solution for your problem will look like without using optimization? If the answer to either of these questions is "yes", you may want to consider a different problem for your project.

- Are there published online demos that have already done what you are trying to do? If so, you may want to consider a different problem for your project, or at least explore the problem in depth that goes beyond what is published online. (For example: I don't think optimizing the selection of a fantasy basketball roster makes a great project.)

- Are there other software packages, Python libraries, etc. that have a pre-packaged implementation of the optimization model you want to solve? If the answer to this question is "yes", you may want to consider a different problem for your project. (For example: I don't think a Pyomo implementation of a common machine learning model makes a great project for a course in optimization. A more interesting project might be to implement a customized machine learning model, using optimization solvers, and then compare against a pre-packaged implementation of common models.)

- Is it possible to obtain an interesting dataset for your problem with a reasonable amount of effort? If the answer to this question is "no", you may want to consider a different problem for your project.

Whatever you choose, I strongly recommend each of you to discuss your project topic with me before investing a significant amount of time in completing it.

4 Define the Problem

Before modeling your problem, you should define and scope your problem. That is, you should be able to explain the motivation and/or application for your problem and make the idea of the problem as clear as possible without math.

5 Formulate an Optimization Model

Formulate a mathematical optimization model for the problem you described in Section 4. The model should be stated in general terms (i.e., no input data specified) so that any problem instance could be solved using your formulation. Clearly explain the idea of the model, and make sure you are also able to answer questions about what each component of the model does.

It is understood that you may have to make some assumptions and/or simplifications along the way to ensure you can populate and solve the model. Be sure to state and justify any assumptions and/or simplifications you have made.

6 Gather and Synthesize the Data

How will you populate the parameters of your model? Be sure to cite the sources of your data, explain what information is contained in the data, and explain in detail how the data is used to populate the parameters of your model. Within reason, it is expected that teams will use high-quality data sources that are of a reasonable scale. If reasonable for your problem, you should provide visualizations to assist in understanding the data.

7 Solve and Analyze the Model

Implement and solve the model(s) using an appropriate solver. Communicate and interpret the results within the context of the original problem. Are the results sensible and meaningful? Does the optimized solution improve upon benchmark alternatives that you could have evaluated without optimization or by using a simpler optimization model? Are there generalizable insights about the structure of an optimal solution? How would the results change in response to a change in the parameters? You don't necessarily have to answer all of these questions; however, you should be seeking out opportunities to learn more about your problem through experimentation with your optimization model. Whatever directions you choose, you may need to solve a number of instances of the problem to gain a thorough understanding of your problem.

If reasonable for your problem, you should provide visualizations of model output to support your analysis.

8 Deliverables

A. During class on April 21, April 23, April 28, or April 30 each team will deliver a 10-minute presentation to communicate the results of the project. The presentation must convey:

- The background and motivating information required for the audience to understand the problem and its importance.
- The underlying idea of the optimization model. (You can *show* the complete mathematical formulation in your slides; however, I suggest focusing oral presentation on the idea of what the model is doing.)
- A summary of the data used to populate and solve the model.
- A summary and discussion of the results obtained from solving the model.

In preparing for the presentation, students must adhere to the following rules:

1. Presentations are expected to last at least 9 minutes but no more than 11 minutes.
2. Slides should be thoughtfully prepared to convey the key ideas/messages (What is the problem? What is the idea of the optimization model? What is the data? What do the results say and why are they meaningful?) as clearly as possible considering the time limitations. I strongly recommend trying to support your slides with visual evidence to the extent possible, and reducing the amount of text/clutter that may be difficult for the audience to process. (For more suggestions and example presentations, I suggest reviewing the resources at https://www.assertion-evidence.com/).

3. The first slide must include the following information:
    a. Project title
    b. Students' full names
    c. Date of the presentation

4. Each subsequent slide must include the slide number in the footer.

5. The final slide should include 3-4 key takeaways from the presentation.

B. Each team should submit a word document that describes the following:

1. the mathematical formulation of the optimization model (e.g., defining and explaining decision variables, parameters, objective, constraints)

2. how the data was collected and used to populate parameter values for the model.

3. (optional) If you completed any analysis that did not appear in your final presentation, please document it here. By providing this documentation, you may be able to better demonstrate the rationale for your final model and results. This documentation can also demonstrate learning on the course project.

4. (optional) If you ran into challenges that limited your progress in solving the problem, explain what you would do differently (or in addition to what you have already done) if you were to continue working on the problem. Be as specific as possible in answering these questions. Within the context of this discussion, this section should provide a sense of what you learned while completing the course project.

This document does not need to be a formal report; rather, it should help someone familiar with optimization to understand the details of your model and analysis.

C. Each student must submit all data and share a link to the code used to build and solve the model. Each student should also submit a final copy of the slides used for the presentation. These materials must be uploaded to Blackboard before the project deadline.

9 Intermediate Deliverables

By March 13, it is expected that each group will have prepared a draft of deliverables B(1) and B(2). Each team *must* schedule one 20-minute meeting with Dr. Sullivan and the TA between March 2 and

March 13 to show their progress on this deliverable and discuss a plan for completing the remainder of the project. I will award some bonus credit to teams who complete this milestone no later than March 6.

10 Expectations for Teamwork

All students are expected to work as effective team members. It is expected that the students within a team may focus more on different parts of the project; however, it is also expected that both team members will review, critique, and ultimately approve all parts of the project (including model formulation, implementation, data collection, and all deliverables) before it is submitted.

11 Rubric

Each team's project will be graded according to the following rubric:

Category (Points Possible)	**Unsatisfactory (0 – 4)**	**Needs Improvement (5 – 6)**	**Adequate (7 – 8)**	**Excellent (9 – 10)**	**Grade**
Define/scope the problem		Both of {motivation, problem idea} are unclear	One of {motivation, problem idea} is unclear or not compelling	Both of {motivation, problem idea} are clear and compelling	
Model formulation		Numerous errors with correctness, not stated generally	Few errors, stated in general terms	Correct, stated in general terms	
Gather and synthesize the data		Lacking in quantity/quality or not relevant to the problem	Relevant data but lacking in quantity, errors in citation	Thorough and relevant to the problem, appropriately cited	
Solve and analyze the model		One instance solved successfully, no significant insights or observations drawn	Fewer instances solved, provides basic observations about solutions	Instances solved correctly, provides significant insights into the specific problem	
Presentation content		Factual errors, inaccurate or incomplete descriptions, conclusions not justified by analysis	Conclusions justified by supporting analysis, but descriptions contain some inaccuracies	Strong conclusions justified by supporting analysis, all descriptions accurate	
Quality of slides		Problems with organization, assertions, and/or evidence that detract from communicating the main ideas	A few problems with organization, assertions, and/or evidence, but effectively communicates the main ideas	Assertions and evidence are strong, follows a logical organization	
Delivery of presentation		Problems with multiple items in the "excellent" list	Problems with some of the items in the "excellent" list, but overall delivery still mostly effective	Confident/ enthusiastic/ knowledgeable, organized and effective with transitions, well-rehearsed but extemporaneous	
Creativity		Directly applies well-known models	Attempts to customize ideas/ models/ datasets/ experiments/ etc. based on the needs of the specific project	Successfully customizes ideas/ models/ datasets/ experiments/ etc. based on the needs of the specific project	
Demonstrates learning through completing the project		Tries something that is a relatively straightforward extension of class concepts/models	Tries something that feels safe; otherwise, tries something uncertain without substantial success or reflection	Tries something ambitious or uncertain; is successful or provides thoughtful reflection based on attempt	